Final Practical – Multi-Stakeholder Presentation & Portfolio Guide

Presenter: Shri Sanjaykumar V (M.Tech Integrated in Software Engineering, VIT Vellore – CGPA: 9.12/10.0)
Audience Tracks:
1. Mentors (*Focus: Engineering Growth, Code Review Responsiveness, Clean UI/UX & End-to-End Polish*)
2. Team Members (*Focus: Modularity, REST API Contracts, Pydantic Schemas, Testing & Git Workflow*)
3. Technical Panel (*Focus: 7-D Isolation Forest Math, Perturbation XAI, Pre-Inference PII Shield, Complexity & Scalability*)

1. Tailored Presentation Strategy by Audience (12–15 Minute Structure)

Track A: Presenting to Mentors (3–4 Minutes)

- Narrative Hook: Connect my real-world experience as Purchase Manager for VIT's Riviera & Gravitas festivals (managing INR 60+ lakhs in purchase orders and vendor settlements) to why I built FinDoc-AuditEngine v3.0.
- Key Evolution to Highlight (Day 26 -> Day 30):

1. Code Review Refactoring (Day 28): Eliminated hardcoded configs, added custom exception hierarchy (FinDocAuditError), and enforced PEP 8 modularity.
2. Institutional Editorial UI & 100% Dynamic Execution (Day 29–30): Replaced generic dark-mode templates with an institutional light accounting grid (Plus Jakarta Sans + IBM Plex Mono tabular numerals) and re-engineered collectLineItemsFromDOM() so every single SKU rate, quantity, and dock rejection is dynamically computed from live user inputs.

- Live Action: Click "Run All 5 Scenarios (Batch)" or "Random Invoice" in the live workbench at http://127.0.0.1:8000 to demonstrate instant dual-currency (1 USD = 86.50 INR) reconciliation.

Track B: Presenting to Team Members (3–4 Minutes)

- Focus Areas: Developer ergonomics, clean separation of concerns (app/core, app/services, app/routers), and zero-regression CI/CD.
- Key Engineering Patterns to Show:

1. Strict API Contracts (app/models.py): Pydantic v2 @model_validator(mode="after") rejects mathematically inconsistent invoices (HTTP 422) before they ever reach business logic.
2. Observability Middleware (app/main.py): Every HTTP response carries X-Correlation-ID and X-Process-Time-Ms, and every audit response embeds a StageTimingTelemetry breakdown (pii_scrub_ms, reconciliation_ms, ml_inference_ms, governance_routing_ms).
3. 30-Test Automated PyTest Suite (tests/): Runs in < 5 seconds via python -m pytest tests/ -v and GitHub Actions (.github/workflows/ci.yml).

Track C: Presenting to the Technical Panel (5–7 Minutes + Q&A)

- Focus Areas: Algorithmic depth, mathematical rigor, security/privacy architecture, and production scaling.
- Deep Technical Talking Points:

1. Why Hybrid Rules + Unsupervised ML? Rules (ReconciliationMatcher) have 100% precision on known contract violations (ΔP_i, $Q_{\text{net}, i}$), while our 7-D StandardScaler + 200-tree IsolationForest (AnomalyScorer) catches zero-day statistical anomalies (such as $49,500 round-dollar offshore invoices just under the $50k CFO limit with Net-1 rush terms) even when the PO matches.
2. How Real-Time Perturbation XAI Works: Instead of returning an opaque anomaly score, assess_risk() computes the marginal risk drop $\Delta R_k(\mathbf{x}) = \max(0, R(\mathbf{x}) - R(\mathbf{x}^{(k \leftarrow \mu_k)}))$ when each feature k is clamped to its historical normal baseline μ_k. Because $D=7$ is compact, all 8 forward passes execute in ~7.9 ms on CPU without external explainer dependencies.
3. Pre-Inference Responsible AI Gate: FinancialPrivacyScrubber executes at Stage 1 (~0.3 ms), replacing PAN, Aadhaar, Bank Account, Credit Card, Phone, and Email tokens with deterministic surrogates ([PAN_01], [EMAIL_02]) so raw PII never enters logs or model vectors.

2. Live Technical Discussion – Top 5 Panel Rapid-Fire Prompts

Panel Question	Concise, High-Impact Engineering Answer
"Show us that your UI and ML model are truly dynamic and not returning fixed preset outputs."	*"In the live workbench (http://127.0.0.1:8000), I can edit any line item's Invoice Rate vs. PO Contract Rate or Dock Rejected quantity, or click Random Invoice. As soon as I change SKU #1's Invoice Rate from $2,100 to $2,450 and add 1 rejected dock unit, ReconciliationMatcher computes the exact dollar variance ($900.00), and AnomalyScorer dynamically recalculates the 7-D Isolation Forest score (97.7/100) and Perturbation XAI bars (Unit Rate Markup Ratio: +39.3 pts, Dock GRN Shortage Ratio: +31.2 pts)."*
"Why did you apply $\ln(1 + \text{Total}_{\text{USD}})$ and StandardScaler before IsolationForest?"	*"While tree splits are invariant to monotonic scaling of a single feature, IsolationForest selects features uniformly at random and draws split thresholds uniformly between feature min and max. Compressing heavy-tailed invoice amounts with $\ln(1+x)$ and normalizing all 7 features via StandardScaler prevents extreme dollar outliers from masking subtle multi-dimensional

interactions."* |
"How do you prevent floating-point rounding bugs in dual-currency USD/INR financial math?"	*"We round line-item products to 2 decimal places (round(qty * price, 2)), enforce a SUBTOTAL_TOLERANCE = 0.05 ($0.05) in both Pydantic v2 validation and 3-way reconciliation, and compute INR deterministically at round(usd * 86.50, 2)."*
"What happens if a vendor submits an unstructured email memo instead of structured JSON?"	*"Our POST /api/v1/audit/parse-raw endpoint runs FinancialPrivacyScrubber.scrub_text() first–redacting PAN, Aadhaar, account, phone, and email tokens–and extracts the Invoice ID, PO Number, Vendor Name, Jurisdiction, and USD Amount directly into the structured workbench."*
"How would you scale this from a single FastAPI node to 10,000 invoices per minute?"	*"Every service class (FinancialPrivacyScrubber, ReconciliationMatcher, AnomalyScorer, ApprovalRouter) is stateless at request time. We scale horizontally behind Gunicorn/Uvicorn workers on Kubernetes, move AUDIT_LEDGER to partitioned PostgreSQL (asyncpg) + Redis idempotency cache, and vectorize the 8 XAI perturbation vectors into a single (8, 7) NumPy batch call to decision_function()."*

3. Master Portfolio Index (GitHub Repositories)

1. Primary Internship & Capstone Repository:
https://github.com/Shri-Sanjaykumar/Linkific-Tasks

- [Day-28/](https://github.com/Shri-Sanjaykumar/Linkific-Tasks/tree/main/Day-28) – PR Workflow, Code Review Report & Refactored FinDoc-AuditEngine
- [Day-29/](https://github.com/Shri-Sanjaykumar/Linkific-Tasks/tree/main/Day-29) – Institutional Editorial Light UI, 10 Technical Interview Q&As, 12-Slide Deck & Demo Suite
- [Day-30/](https://github.com/Shri-Sanjaykumar/Linkific-Tasks/tree/main/Day-30) – Final Capstone Portfolio Hub, 2–3 Page Reflection Report, Updated Resume (PDF + MD), LinkedIn Guide, 90-Day Roadmap & 30/30 PyTest Suite

2. Structured Python AI/ML Mirror Repository:
https://github.com/Shri-Sanjaykumar/AI-ML-Internship